

Calibration Program Setup Checklist

A step-by-step guide to building a compliant calibration program from scratch

- ✓ Instrument inventory and classification checklist
- ✓ Calibration interval optimization guide
- ✓ Lab selection and qualification process
- ✓ Audit-ready documentation and record-keeping

Step 1: Instrument Inventory

Build Your Master Instrument List

- ☐ Identify all measuring and test equipment (M&TE) in your facility
- ☐ Assign a unique identification number to every instrument
- ☐ Record: instrument type, manufacturer, model, serial number, location, custodian
- ☐ Classify each instrument as Critical, Important, or Reference-Only (see classification table below)
- ☐ Determine which instruments require calibration (affect product quality or are used for accept/reject decisions)
- ☐ Identify instruments that do NOT require calibration (indication-only, shop tools, non-measurement equipment)
- ☐ Label all calibrated instruments with a unique ID, calibration date, and due date
- ☐ Label all non-calibrated instruments clearly (e.g., "NOT CALIBRATED — FOR REFERENCE ONLY")

Classification	Definition	Calibration Required?	Accredited Required?
Critical	Used for final product acceptance, safety measurements, regulatory compliance	Yes — mandatory	Yes — ISO 17025 accredited
Important	Used in-process for quality-relevant measurements, not final acceptance	Yes — mandatory	Recommended; traceable-only may be acceptable
Reference-Only	Used for indication, estimation, or non-quality purposes	No — label as exempt	N/A
Fixed / Built-In	Integrated into equipment (e.g., oven display, CNC readout)	Depends — assess impact on product quality	Depends on application

Every instrument in your facility must be either calibrated and labeled, or clearly identified as exempt.

Auditors look for unlabeled instruments first — an unlabeled micrometer on a workbench is an instant nonconformance.



Step 2: Set Calibration Intervals

How Often Should You Calibrate?

Starting-Point Intervals

Instrument Type	Recommended Starting Interval	Factors That Shorten	Factors That Lengthen
Calipers, micrometers	12 months	Heavy use, harsh environment	Light use, lab environment
Gauge blocks	12-24 months	Frequent handling	Seldom used reference set
Pressure gauges	6-12 months	Vibration, pressure cycling	Static application
Torque wrenches	6-12 months	Heavy daily use	Infrequent use
Digital multimeters	12 months	Extreme temperatures	Lab environment, light use
Thermocouples / RTDs	6-12 months	High temperature cycling	Narrow range use
Scales / balances	6-12 months	Heavy traffic, dust	Clean lab environment
Oscilloscopes	12 months	Age, frequency of use	Recent purchase, light use

Interval Optimization Methods

Method	How It Works	Best For
Calendar-based (fixed)	Calibrate every X months regardless of use	Simple, easy to manage; conservative
Usage-based	Calibrate after X uses or measurements	Instruments with usage counters or logs
Statistical (NCSL RP-1)	Analyze as-found data to optimize intervals — extend if consistently in-tolerance	Mature programs with 3+ cycles of data

Method	How It Works	Best For
Risk-based	Assign intervals based on criticality, usage, stability history, and consequence of failure	ISO 10012 and advanced programs

Start conservative, then optimize.

Begin with manufacturer-recommended intervals (or 12 months if none specified). After 3-5 calibration cycles, review as-found data. If an instrument has been in-tolerance for 5 consecutive cycles, extend the interval. If it's been out-of-tolerance twice, shorten it.

Step 3: Select and Qualify Calibration Labs

Checklist for lab selection

- ☐ Identify calibration disciplines needed (dimensional, electrical, pressure, temperature, etc.)
- ☐ Search for accredited labs covering your specific scope (use CalLedger, A2LA directory, NVLAP directory)
- ☐ Verify ISO/IEC 17025 accreditation AND scope of accreditation covers your parameters and ranges
- ☐ Confirm BMC (Best Measurement Capability) provides adequate TUR (4:1 minimum) for your tolerances
- ☐ Request and evaluate sample calibration certificates for completeness
- ☐ Compare turnaround times (standard vs. rush), pricing, and service options (ship-to-lab vs. on-site)
- ☐ Request client references, preferably from your industry
- ☐ Evaluate logistics: location, shipping procedures, pickup/delivery, instrument handling
- ☐ Obtain Certificate of Insurance (for labs handling high-value equipment)
- ☐ Document your lab qualification decision (required for most quality system audits)

Approved Vendor Qualification Checklist

Verification Item	Status
ISO/IEC 17025 accreditation verified	[] Complete
Scope covers required parameters and ranges	[] Complete
BMC provides adequate TUR (4:1 minimum)	[] Complete
Sample certificates reviewed and acceptable	[] Complete
Proficiency testing participation confirmed	[] Complete

Verification Item	Status
OOT notification procedure documented	[] Complete
References checked	[] Complete
Pricing and terms agreed	[] Complete
Lab added to Approved Vendor List	[] Complete

Step 4: Documentation & Record-Keeping

Building Audit-Ready Calibration Records

Required Documentation

- ☐ **Master Instrument List** — Complete inventory of all M&TE with calibration status
- ☐ **Calibration Schedule** — Upcoming due dates for all instruments, managed in software or spreadsheet
- ☐ **Calibration Procedures** — Written procedures for any calibrations performed in-house
- ☐ **Calibration Certificates** — Filed and indexed for every external calibration (retain minimum 2 calibration cycles, longer for regulated industries)
- ☐ **As-Found / As-Left Records** — Before and after adjustment data for every calibration
- ☐ **Out-of-Tolerance Reports** — Investigation and impact assessment when instruments are found OOT
- ☐ **Approved Vendor List** — Qualified calibration labs with accreditation verification records
- ☐ **Instrument History Files** — Cumulative record of all calibrations, adjustments, and repairs for each instrument
- ☐ **Calibration Labels / Stickers** — Applied to every calibrated instrument showing ID, calibration date, due date

Out-of-Tolerance Response Procedure

Step	Action
1	Quarantine the OOT instrument immediately
2	Notify quality and affected departments
3	Review all measurements made with this instrument since the last known good calibration
4	

Step	Action
	Assess impact on product quality — determine if any product must be recalled, re-inspected, or re-tested
5	Document the OOT condition, investigation, and corrective action
6	Have the instrument repaired and recalibrated, or replace it
7	Update calibration interval if the OOT event suggests the current interval is too long

OOT events are audit gold (or audit nightmares).

When an instrument is found out of tolerance, you must be able to show that you investigated the impact on all measurements made since the last calibration. An OOT event with a documented, thorough investigation demonstrates a mature quality system. An OOT event with no investigation is a major nonconformance.

Step 5: Calibration Software & Management

Tools for Managing Your Calibration Program

Software Comparison

Tool	Type	Best For	Approximate Cost
Excel / Google Sheets	Spreadsheet	Small programs (<50 instruments), startups	Free
Fluke MET/CAL + MET/TEAM	Enterprise CMMS	Large metrology departments, automated calibration	\$5,000-\$50,000+
GAGetrak	Dedicated calibration management	Mid-size programs, 50-5,000 instruments	\$2,000-\$15,000/yr
Calibration Control (Ape Software)	Cloud-based	Small to mid-size, web access	\$1,000-\$5,000/yr
Blue Mountain RAM	Enterprise asset management	Large facilities, integration with ERP/CMMS	\$10,000-\$100,000+
Master Control	QMS with calibration module	Pharma/medical device (FDA-regulated)	\$20,000-\$100,000+/yr
IndySoft	Calibration management	Mid to large programs, lab and field	\$5,000-\$25,000+

Minimum software requirements for audit compliance

- ☐ Unique instrument identification and searchable database
- ☐ Automatic calibration due-date tracking and overdue alerts
- ☐ Calibration history per instrument (as-found, as-left, adjustments, OOT events)
- ☐ Approved vendor management (lab accreditation status and scope tracking)
- ☐ Certificate storage and retrieval

☐ Audit trail / change log (who changed what, when)

Start with a spreadsheet, migrate to software when you hit 100 instruments.

A well-designed Excel tracker works perfectly for small programs. When you outgrow it (missed due dates, lost certificates, audit prep taking days instead of hours), invest in dedicated calibration management software.

Audit Preparation Quick Reference

Preparing for Quality System Audits — Calibration Section

What Auditors Check

Audit Check	What They're Looking For	How to Prepare
1. Calibration status of M&TE on the production floor	Every instrument has a current label showing cal date, due date, and ID	Walk the floor and check every instrument weekly
2. Overdue instruments	Any instruments past their calibration due date	Run your overdue report daily; quarantine overdue instruments immediately
3. Calibration certificates on file	Certificates for every external calibration, complete with traceability info	Indexed filing system (physical or digital); instant retrieval
4. Approved vendor list with accreditation verification	Labs verified as accredited, scope checked	Annual re-verification of lab accreditation status
5. OOT investigation records	Documented impact assessment for any OOT findings	Completed investigation form for every OOT event
6. In-house calibration procedures	Written procedures for any calibrations done internally	Procedures per ISO/IEC 17025 or per your QMS
7. Master instrument list accuracy	List matches what's actually on the floor	Annual physical inventory reconciliation
8. Calibration intervals justified	Rational basis for each instrument's interval	Document interval rationale; maintain as-found trending data

Top 3 Audit Findings (in order of frequency)

1. Instruments on the floor with expired or missing calibration labels
2. Calibration certificates missing or incomplete (no traceability, no uncertainty)
3. No investigation or impact assessment for out-of-tolerance findings

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1. ****Printability matters.**** Quality managers and metrology teams print these for reference binders and audit prep. Avoid large solid color areas on interior pages (waste ink). Reserve full-bleed backgrounds for cover and back cover only. 2. ****Tables are the star.**** These guides live or die on their reference tables. Make tables scannable, well-spaced, and easy to read at 100% zoom on screen AND w

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